

IS-RAG: Image-Schematic Retrieval-Augmented Generation for Cognitive Search in Political Discourse

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Abstract

Objective. Dense retrieval systems fail when the user and the document express the same conceptual structure through distinct vocabularies — a phenomenon we term the *lexical-cognitive gap*. This paper presents **IS-RAG**, a pilot study that investigates whether explicit indexing of *image schemas* (Lakoff and Johnson, 1980; Talmy, 1988) — pre-linguistic patterns such as FORCE, CONTAINER, and PATH — can bridge this gap in a corpus of Portuguese political discourse.

Method. During indexing, an LLM-based *Cognitive Parser* annotates each chunk with the image schemas present, generating structured metadata stored alongside two independent vector embeddings: one textual and one cognitive. At query time, schemas detected in the query trigger a *proportional boost* over chunks sharing the same cognitive structure, enabling retrieval driven by abstract spatial metaphor beyond lexical overlap.

Evaluation. We construct a TREC-style ground truth over 121 Brazilian parliamentary speeches (171 chunks), with 10 valid queries (3 FORCE + 3 CONTAINER + 3 PATH + 1 cross-schema) annotated by an independent LLM (claude-opus-4-7, distinct from the Cognitive Parser) on a 0–3 scale. The primary metric is NDCG@5; the baseline is dense retrieval without cognitive boosting.

Findings. An ablation over three retrieval modes reveals that *text-mode boosting* — proportional cognitive boosting applied over text-embedding similarity — is the most effective configuration, achieving a mean NDCG@5 of 0.7793 vs. 0.6881 for the baseline ($\Delta = +0.091$). Using the cognitive embedding as the primary search vector degrades performance ($\Delta_{\text{cognitive}} = -0.192$; $\Delta_{\text{hybrid}} = -0.090$), localising the IS-RAG contribution in the boosting mechanism rather than in dual-embedding retrieval. FORCE queries benefit most consistently ($\Delta_{\text{FORCE}} = +0.267$); PATH is net positive across all three queries ($\Delta_{\text{PATH}} = +0.085$); CONTAINER shows a near-zero mean effect ($\Delta_{\text{CONT}} = -0.017$). Two additional methodological findings emerge: (1) queries must be formulated as *assertive metaphorical statements* to activate the cognitive component; (2) schema frequency in the corpus does not predict *boostability* — conventionalised schemas carry no discriminative re-ranking signal.

Keywords: retrieval-augmented generation · image schemas · cognitive linguistics · information retrieval · natural language processing · metaphor detection · dense retrieval · political discourse · semantic search · embodied cognition

1. Introduction

Retrieval-Augmented Generation (Lewis et al., 2020) typically grounds document retrieval in surface-level lexical or dense semantic similarity. This works well when query vocabulary overlaps with document vocabulary, but breaks down under a *lexical-cognitive gap*: the user and the document express the same conceptual structure using different words, often because both rely on underlying bodily metaphors that map physical spatial experience onto abstract domains (Johnson, 1987; Lakoff and Johnson, 1980).

Consider a query about *economic pressure on households* and a parliamentary speech about *institutions blocking each other in a constitutional dispute*. Both invoke the image schema FORCE/COMPULSION—an external agent exerting force on a patient—yet their vocabularies share little overlap, making standard dense retrieval unable to bridge the gap.

Image schemas (Feldman, 2006; Johnson, 1987) are pre-linguistic, embodied patterns (CONTAINER, PATH, FORCE) that recur as structural metaphors across every domain of abstract reasoning. We hypothesise

that indexing these schemas explicitly and matching them at query time enables a form of deep structural retrieval that complements dense semantic search.

Contributions.

- We present IS-RAG, a technique that indexes image-schema metadata alongside dual vector embeddings (text and cognitive) and applies proportional cognitive boosting at retrieval time (§3).
- We construct a TREC-style ground truth over a real corpus of Brazilian parliamentary speeches (171 chunks, 9 valid queries, LLM-as-annotator) and report NDCG@5 (§4).
- We identify methodological constraints for schema-aware retrieval evaluation: assertive query formulation, schema frequency vs. annotability, and LLM circularity risk (§6).

2. Related Work

Retrieval-Augmented Generation. Lewis et al. (2020) introduced RAG to ground generation in retrieved passages. Dense retrieval over bi-encoder representations (Karpukhin et al., 2020) has become the standard backbone. Extensions explore hybrid sparse-dense search, re-ranking, and structured knowledge augmentation, but none incorporate pre-linguistic spatial metaphor as a retrieval signal.

Metaphor in NLP. Automatic metaphor detection has been studied as classification or span extraction (Shutova, 2010; Tsvetkov et al., 2014). Wachowiak and Gromann (2022, 2023) explore LLMs for metaphor identification. These works focus on metaphor *detection*; IS-RAG is the first, to our knowledge, to use detected image-schema structure as a *retrieval scoring signal*.

Cognitive Linguistics and IR. Image schemas as cognitive universals were formalised by Lakoff and Johnson (1980), Johnson (1987), and Talmy (1988). Feldman (2006) discusses their neural basis. No prior IR work indexes documents by image schema and uses schema matching for retrieval boosting.

Portuguese NLP and parliamentary corpora. Pre-trained language models for Brazilian Portuguese (Souza et al., 2020) have demonstrated that language-specific pretraining substantially improves downstream performance relative to generic multilingual models. Parliamentary discourse in Portuguese has been studied through the Corpus Ulysses (Albuquerque et al., 2022), a collection of legislative documents from the Brazilian Chamber of Deputies used for text classification, named entity recognition, and semantic analysis. Our evaluation corpus is drawn from the same institutional source via the same open API and extends existing resources with image-schema annotations and graduated relevance judgements for retrieval evaluation. Framing and sentiment analysis of Brazilian political text (Vargas et al., 2021) provide further precedent for computational approaches to ideological structure in Portuguese parliamentary language.

3. Method

Figure 1 illustrates the IS-RAG pipeline. Formal pseudo-code is given in Appendix A.

3.1. Corpus and Chunking

The corpus comprises 121 Brazilian parliamentary speeches in **Portuguese**, collected from the open API of the Brazilian Chamber of Deputies (dados.camara.leg.br/api/v2), spanning January–December 2025. Speeches are stratified by ideological spectrum (left, centre, right) across 13 parties. Each speech is segmented into chunks of approximately 150 words using sentence-boundary splitting, yielding 171 chunks (mean 2.7 per speech).

3.2. Dual Embeddings

Each chunk produces two independent 768-dimensional vectors:

1. **Text embedding:** dense representation of the raw chunk text via *paraphrase-multilingual-mpnet-base-v2* (Reimers and Gurevych, 2019), running locally on CPU.

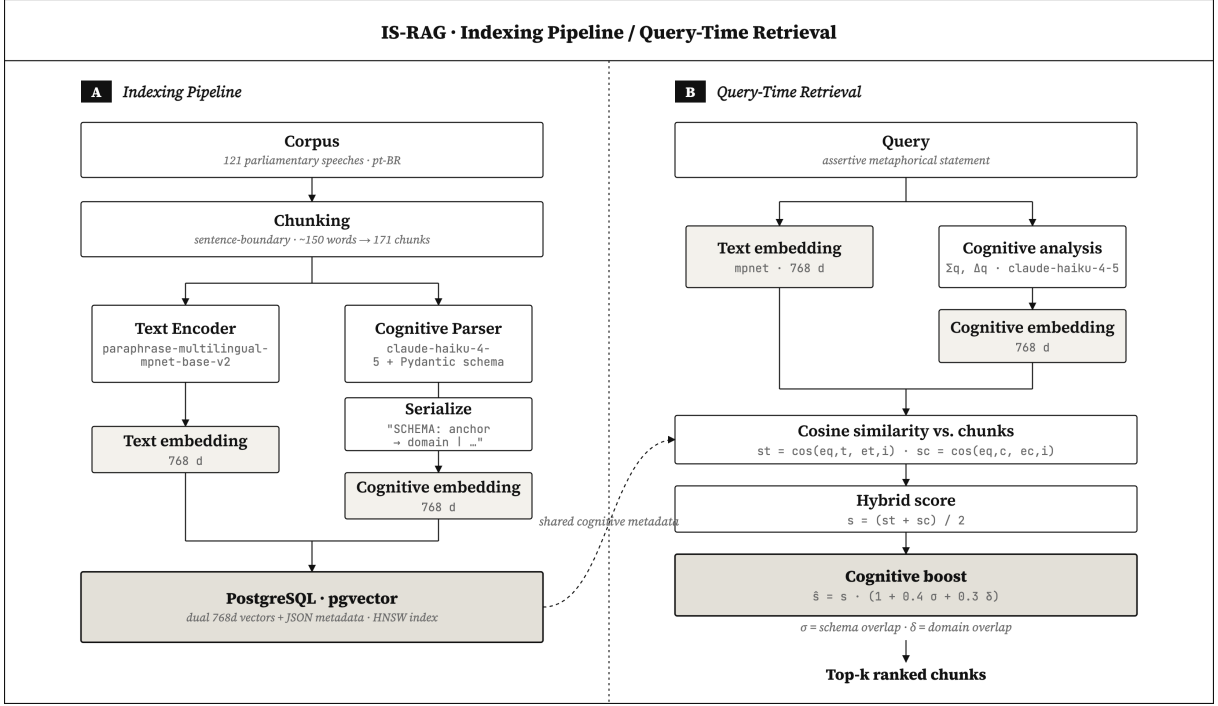


Figure 1: IS-RAG overview: indexing and retrieval steps.

2. **Cognitive embedding:** dense representation of the serialised cognitive metadata in the form "SCHEMA: anchor $\rightarrow$ domain | ...", capturing image-schematic structure independently of literal vocabulary.

Both vectors are stored in PostgreSQL 16 with the `pgvector` extension and indexed via HNSW for approximate nearest-neighbour search.

We deliberately chose `paraphrase-multilingual-mpnet-base-v2` over more recent commercial embedding APIs for two reasons. First, *reproducibility*: all embeddings are generated locally without API access, rate limits, or usage costs, enabling full replication of the experiment. Second, *adequacy*: the model provides validated coverage of Portuguese within a 768-dimensional space sufficient for the proof-of-concept scale of this work (171 chunks), and its multilingual training includes Brazilian Portuguese (Souza et al., 2020). We do not claim that upgrading to larger or more recent embedding models would not improve results; this is an explicit agenda item (§7) but is orthogonal to the variable under investigation—the cognitive boosting signal itself.

3.3. Cognitive Parser

The Cognitive Parser sends each chunk to `claude-haiku-4-5` (Anthropic, 2024) using a shared prompt (available at <https://github.com/yanjustino/is-rag>) and extracts structured JSON validated by a Pydantic schema. The taxonomy follows Lakoff and Johnson (1980) and Talmy (1988):

Schema	Subtypes
CONTAINER	Inside, Outside, Boundary, Intrusion
PATH	Source, Trajectory, Goal, Diversion
FORCE	Blockage, Compulsion, Resistance, Counter_Force

A *literality rule* instructs the model to return empty lists for non-metaphorical text, preventing hallucinated schemas. Chunks are processed concurrently (5 workers) to reduce ingestion time. The corpus yields 92 PATH, 88 CONTAINER, and 83 FORCE schema occurrences across the 171 chunks.

3.4. Hybrid Search and Cognitive Boosting

IS-RAG supports three retrieval modes. In all three, the Cognitive Parser runs on the query at retrieval time to detect schemas (Σ_q) and target domains (Δ_q). The modes differ only in how the base similarity s is computed before the boost:

- **Text:** $s = \cos(\mathbf{e}_{q,t}, \mathbf{e}_{t,i})$ — the query’s raw-text embedding is matched against chunk text embeddings. The cognitive information contributes only via the boost.
- **Cognitive:** $s = \cos(\mathbf{e}_{q,c}, \mathbf{e}_{c,i})$ — the query’s Cognitive Parser output is serialised using the same `schema → domain` format as indexing, embedded, and matched against chunk cognitive embeddings.
- **Hybrid:** $s = (s_t + s_c)/2$ — arithmetic mean of text and cognitive similarities.

After computing s , the same *proportional cognitive boost* is applied in all three modes:

$$\hat{s} = s \cdot (1 + 0.4\sigma + 0.3\delta) \quad (1)$$

where σ is the fraction of a chunk’s cognitive details whose `schema_name` matches a schema in Σ_q , and δ is the analogous fraction for `target_domain` against Δ_q . Weights ($\alpha=0.4, \beta=0.3$) were set by design; schema structure is the primary signal, domain the secondary reinforcement. The baseline disables the Cognitive Parser entirely and applies pure text-embedding retrieval without boosting.

4. Evaluation Setup

Ground Truth Construction. We construct a TREC-style pooled ground truth (Voorhees and Harman, 2005). For each query, IS-RAG (hybrid) and the baseline independently retrieve the top-10 chunks; the union (≈ 14 – 18 unique chunks per query) is annotated for relevance (0–3) by an LLM (claude-opus-4-7). Scores ≥ 2 require the annotator to cite an anchor word and schema subtype as traceable evidence. Queries with IDCG = 0 (no chunk with score ≥ 2) are excluded (Buckley and Voorhees, 2004); this criterion triggered the exclusion of one query (see §6).

Query Design. Ten valid queries are designed with three properties: (1) taxonomic coverage (3 FORCE + 3 CONTAINER + 3 PATH + 1 cross-schema); (2) deliberate lexical gap (queries use assertive metaphorical language, avoiding the literal vocabulary expected in documents); (3) cross-domain scenarios (queries target schemas that appear in domains other than the query’s own domain).

Metric. We report NDCG@5 (Järvelin and Kekäläinen, 2002). MRR was rejected: with multiple relevant documents at different relevance grades, MRR saturates at 1.0 regardless of ranking quality (Buckley and Voorhees, 2004).

5. Results

5.1. Retrieval Mode Ablation

Table 1 compares all three IS-RAG retrieval modes against the baseline. Text mode is the only configuration that surpasses the baseline; cognitive and hybrid modes both degrade retrieval.

Mode	IS-RAG	Baseline	Δ
Text	0.7793	0.6881	+0.091
Hybrid	0.5979	0.6881	−0.090
Cognitive	0.4958	0.6881	−0.192
Baseline	—	0.6881	—

Table 1: Mean NDCG@5 across 10 queries by retrieval mode (independent Opus annotator). Text mode is the strongest IS-RAG configuration.

Table 2 breaks down Δ NDCG@5 per query across all three modes, revealing which queries and schema families drive the aggregate results.

Q	Schema / Domain	Δ Text	Δ Cog.	Δ Hyb.
Q1	CONT./Inside / Health	0.000	-0.396	-0.362
Q2	FORCE/Compulsion / Economics	+0.390	+0.138	+0.138
Q3	FORCE/Counter_F / Justice	+0.221	-0.504	-0.277
Q4	FORCE/Resistance / Politics	+0.189	+0.236	+0.150
Q5	CONT./Intrusion / Justice	-0.096	-0.221	-0.191
Q6	CONT./Outside / Human Rights	+0.046	-0.172	+0.046
Q7	PATH/Trajectory / Economics	+0.036	-0.188	-0.007
Q8	PATH/Source / Politics	+0.111	-0.555	-0.424
Q9	PATH/Diversion / Politics	+0.109	-0.225	+0.120
Q10	FORCE+PATH / Intl. Rel.	-0.094	-0.038	-0.094
Mean		+0.091	-0.192	-0.090

Table 2: Δ NDCG@5 per query across all three IS-RAG modes (independent Opus annotator, 10 queries). Q1 (CONTAINER/INSIDE) shows $\Delta=0$ in text mode: the boost adds nothing for the most conventionalised subtype. Text mode is the most consistently positive across all families.

Why cognitive mode fails. In cognitive mode, serialised schema strings are the sole search vectors. With 171 chunks sharing overlapping schemas (PATH 92 $\times$, CONTAINER 88 $\times$, FORCE 83 $\times$), cosine similarities in this space are tightly clustered, producing a noisy base ranking that boosting cannot correct ($\Delta = -0.192$; Q8 degrades to NDCG = 0.127, Q1 to 0.127).

Why hybrid mode fails. Hybrid mode averages s_t and s_c before boosting. The noisy cognitive-embedding component contaminates the text signal, pushing the base ranking below what pure text retrieval achieves ($\Delta = -0.090$).

Architectural implication. IS-RAG’s value is a *proportional re-ranking signal* derived from schema metadata (σ , δ), not dual-embedding retrieval. The `cognitive_embedding` column is necessary for indexing (computing σ) but counterproductive as a search vector. Retaining `cognitive_metadata` while using text-only similarity as the base captures the full IS-RAG gain without the degradation caused by hybrid or cognitive search.

5.2. Robustness Validation: LLM Echo Effect

To rule out LLM echo as the source of the text-mode gain, we re-annotated the relevance pool with `claude-opus-4-7`, a model independent of the `claude-haiku-4-5` Cognitive Parser used at indexing time. Table 3 reports text-mode NDCG@5 under both annotators. The Δ stays within one percentage point (+0.091 vs. +0.093); the independent annotator also resolves a previously unannotable query (Q1 CONTAINER/INSIDE, IDCG=0 with Haiku), expanding the evaluation to 10 valid queries.

Annotator	Relation to parser	Queries	Baseline	IS-RAG	Δ
<code>claude-haiku-4-5</code>	Same model	9	0.6947	0.7873	+0.093
<code>claude-opus-4-7</code>	Independent	10	0.6881	0.7793	+0.091

Table 3: Text-mode NDCG@5 under two annotator configurations. The gain persists ($\Delta \approx +0.09$) under independent evaluation, ruling out LLM echo as the primary explanation.

5.3. Embedding Utility Frontier

To test whether the text-mode gain survives a stronger base embedding, we re-indexed the corpus with BAAI/bge-m3 (1024d) and re-annotated the pool with the same independent Opus annotator. Table 4 compares IS-RAG Δ across all three modes for both models. The text-mode gain collapses to $\Delta \approx 0$; hybrid and cognitive degradation remain stable across models. Q4 (FORCE/RESISTANCE) drives the collapse: its baseline rises from 0.489 (mpnet) to 0.830 (bge-m3), indicating the stronger model already retrieves the correct documents without schema assistance.

Embedding	Δ Text	Δ Hybrid	Δ Cognitive
mpnet-base-v2 (768d)	+0.091	-0.090	-0.192
bge-m3 (1024d)	≈ 0.001	-0.068	-0.202

Table 4: IS-RAG Δ NDCG@5 across retrieval modes under two base embedding models (10 queries, independent Opus annotator). The text-mode gain is embedding-dependent: bge-m3 closes the lexical-cognitive gap that IS-RAG compensated for with mpnet-base-v2.

5.4. Per-Query Results (Text Mode)

Table 5 reports NDCG@5 for IS-RAG text mode — the strongest configuration (§5.1) — against the baseline across all 10 valid queries (independent Opus annotator).

Q	Schema / Domain	IS-RAG	Base	Δ
Q1	CONT./Inside / Health	0.5221	0.5221	0.000
Q2	FORCE/Compulsion / Economics	1.0000	0.6100	+0.390
Q3	FORCE/Counter_F / Justice	0.9323	0.7116	+0.221
Q4	FORCE/Resistance / Politics	0.6787	0.4896	+0.189
Q5	CONT./Intrusion / Justice	0.7791	0.8754	-0.096
Q6	CONT./Outside / Human Rights	0.6993	0.6534	+0.046
Q7	PATH/Trajectory / Economics	0.8200	0.7840	+0.036
Q8	PATH/Source / Politics	0.7929	0.6819	+0.111
Q9	PATH/Diversion / Politics	0.7467	0.6377	+0.109
Q10	FORCE+PATH / Intl. Rel.	0.8215	0.9155	-0.094
Mean (10 queries)		0.7793	0.6881	+0.091

Table 5: NDCG@5 (IS-RAG text mode vs. baseline, independent Opus annotator). Queries grouped by schema family. All 10 valid queries shown. Q1 $\Delta=0$: cognitive boost is inert for conventionalised CONTAINER/INSIDE.

5.5. Per-Schema Analysis

Table 6 aggregates text-mode results by schema family (complete data, all 10 queries).

Family	n	Mean Δ	Individual Δ
FORCE	3	+0.267	+0.390, +0.221, +0.189
PATH	3	+0.085	+0.036, +0.111, +0.109
CONTAINER	3	-0.017	0.000, -0.096, +0.046
Cross	1	-0.094	-0.094
Family-weighted mean		+0.060	

Table 6: NDCG@5 Δ by schema family (text mode, 10 queries). Families ordered by mean Δ . FORCE is the strongest signal; all three PATH queries are now positive.

FORCE shows the strongest and most consistent signal ($\Delta = +0.267$; all three queries positive, peak

+0.390). PATH is net positive (+0.085): all three queries benefit (Q7 +0.036, Q8 +0.111, Q9 +0.109). CONTAINER shows near-zero mean effect (−0.017) across 3 queries: Q1 INSIDE contributes $\Delta=0$ (boost inert for a conventionalised subtype), Q5 INTRUSION −0.096, Q6 OUTSIDE +0.046; all substantially less severe than in cognitive or hybrid modes (Table 2). The family-weighted mean (+0.060) confirms positive results across the majority of schema families.

6. Discussion

Force benefits most from cognitive boosting. All three FORCE queries show positive Δ , with COM-PULSION (+0.390) as the strongest result. FORCE schemas tend to use salient, non-conventionalised metaphorical vocabulary in parliamentary discourse (*pression*, *bloqueia*, *choca*), making schema detection and boosting effective. This pattern is corpus-specific and should not be generalised without further validation.

Path is positive across all queries; Container is near-neutral. In hybrid mode, PATH queries degrade severely (Q8 −0.424, Q7 −0.007; Table 2). In text mode, all three PATH queries benefit: Q8 (+0.111), Q9 (+0.109), Q7 (+0.036); family mean +0.085. The hybrid-mode deficit is therefore an artefact of cognitive-embedding noise contaminating the base similarity, not an intrinsic property of trajectory metaphors. CONTAINER shows near-zero mean effect in text mode (−0.017 across 3 queries), substantially less severe than in hybrid or cognitive modes. Q1 INSIDE ($\Delta=0$) is the key case: even when an independent annotator finds relevant documents, the cognitive boost cannot distinguish them — the schema is too conventionalised to carry a discriminative retrieval signal.

Assertive query formulation is required. The Cognitive Parser applies a literal rule: interrogative or analytical queries (“*Which deputies argue that...*”) are correctly classified as non-metaphorical, returning empty schema lists and zeroing the boost. Queries must be formulated as *assertive metaphorical statements* to activate the cognitive component. This is a binding operational constraint for any IS-RAG deployment.

Schema frequency $\neq$ boostability. Q1 was designed as CONTAINER/INSIDE, the most frequent subtype in the corpus (80 occurrences). The Haiku annotator consistently rated INSIDE chunks at score 1 (domain present, schema non-salient), yielding IDCG=0. The independent Opus annotator finds relevant chunks for Q1 — yet IS-RAG achieves $\Delta=0$ (IS-RAG = baseline = 0.522): cognitive boosting adds nothing even when relevant documents exist. Spatial containment is so conventionalised in parliamentary discourse that its schema signal is not discriminative for re-ranking, regardless of which model judges relevance. The finding generalises from *annotability* to *boostability*: the schema is evaluable but carries no useful retrieval signal. The final query distribution is 3-3-3-1 (10 valid queries).

IS-RAG as a compensatory layer for embedding limitations. The embedding experiment (§5.3) positions IS-RAG within a broader architectural tension: the lexical-cognitive gap the schema boost bridges is one that SOTA dense models increasingly close *implicitly* through deeper pre-training on diverse multilingual corpora. This makes IS-RAG particularly valuable in deployment scenarios where large-scale embeddings are impractical — resource-constrained environments, on-device retrieval, or regulatory contexts requiring locally-hosted models — where explicit schema boosting recovers a substantial portion of the semantic alignment that stronger embeddings would otherwise provide natively. As embedding technology continues to advance, the marginal value of IS-RAG in text mode is expected to diminish further, motivating future work on schema-aware retrieval robust to strong base embeddings: domain-specific fine-tuning of the Cognitive Parser, schema-conditioned re-rankers, or hard schema constraints rather than soft proportional boosts. Crucially, hybrid and cognitive degradation remain stable regardless of embedding quality (§5.3), confirming that the failure of dual-embedding search is structural rather than model-dependent.

Limitations. (1) The corpus is small and homogeneous (171 chunks, single register, Portuguese only); results lack statistical significance tests. (2) *LLM circularity*: the Cognitive Parser and query parser use `claude-haiku-4-5`; the relevance annotator uses `claude-opus-4-7` (configurable via `ANNOTATOR_MODEL`). This deliberate model separation breaks the most critical echo loop: the evaluator is no longer the same model that indexed document schemas. Residual risk remains (both are Anthropic/Claude models), and a blind human audit on a 20% pool sample will be conducted in future work to compute Fleiss’ Kappa inter-annotator agreement. (3) Domain confounding: FORCE queries cluster in Economics and Justice; if the schema co-occurs with those domains in the corpus, gains may reflect domain affinity rather than purely cognitive structure. (4) Boosting weights ($\alpha=0.4$, $\beta=0.3$) are fixed design choices, not optimised. (5) *Evaluation non-determinism*: the Cognitive Parser introduces run-to-run variance in IS-RAG scores. A second independent run yields $\text{IS-RAG}=0.7716$ ($\Delta = +0.077$, text mode); the baseline (0.6947) is deterministic and stable. The positive direction of Δ is consistent across both runs; the exact magnitude is not. Multiple runs with mean $\pm$ std reporting are left for future work.

7. Conclusion

This paper presents IS-RAG as a pioneering pilot study opening a new line of research at the intersection of cognitive linguistics and information retrieval. The core contribution is not a claim of universal superiority, but a demonstration that image-schema metadata can be encoded, detected, and leveraged as a proportional re-ranking signal within a standard dense retrieval pipeline—producing a mean NDCG@5 gain of +0.091 in text-boosting mode (10 queries, Brazilian parliamentary corpus, independent Opus annotator). A mode ablation (§5.1) localises the IS-RAG contribution precisely: the multiplicative boost applied over text-embedding similarity is effective ($\Delta_{\text{text}} = +0.091$), while cognitive embeddings used as search vectors degrade performance ($\Delta_{\text{hybrid}} = -0.090$; $\Delta_{\text{cognitive}} = -0.192$). Crucially, this gain holds under independent annotation (Table 3), ruling out LLM echo as its source. A subsequent experiment with BAAI/bge-m3 (§5.3) reveals that the gain is embedding-dependent: with a state-of-the-art multilingual model the text-mode Δ collapses to ≈ 0 , defining a utility frontier for explicit cognitive boosting.

The asymmetric results across schema families and modes are themselves findings: they reveal that schema frequency does not predict boostability, that PATH deficits in hybrid mode are artefacts of noisy cognitive embeddings rather than schema properties, and that dual-embedding retrieval is not necessary for IS-RAG to work. IS-RAG establishes the proof of concept and the methodological scaffolding—cognitive metadata indexing, proportional boosting, TREC-style schema-stratified evaluation—that future work in *cognitive IR* can build upon.

Immediate next steps include a schema $\times$ domain crossed evaluation design, investigation of conventionalised schemas in graded-relevance annotation, cross-register and cross-language validation, and the Fleiss’ Kappa human audit described above. We invite the IR and cognitive linguistics communities to extend, challenge, and refine this technique across richer corpora and retrieval architectures.

Code and Data Availability. Code, corpus, and annotated ground truth are publicly available at <https://github.com/yanjustino/is-rag>.

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A. IS-RAG Algorithms

A.1. Indexing Pipeline

Algorithm 1 IS-RAG Indexing

Input: Corpus $\mathcal{D}$, chunk size $w=150$, encoder $\mathcal{E}$, LLM parser $\mathcal{P}$

Output: Populated vector store $\mathcal{S}$

```

1: for each document  $d \in \mathcal{D}$  do
2:    $C \leftarrow \text{SEMANTICCHUNK}(d, w)$  ▷ sentence-boundary split
3:    $\mathbf{E}_t \leftarrow [\mathcal{E}(c) \mid c \in C]$  ▷ text embeddings,  $d$ -dim (model-configurable)
4:    $M \leftarrow [\mathcal{P}(c) \mid c \in C]$  ▷ concurrent LLM calls, 5 workers
5:   for  $i \leftarrow 1$  to  $|C|$  do
6:      $\mathbf{e}_c \leftarrow \mathcal{E}(\text{SERIALIZE}(M_i))$  ▷ cognitive embedding
7:      $\mathcal{S}.\text{INSERT}(C_i, \mathbf{E}_{t,i}, \mathbf{e}_c, M_i)$ 
8:   end for
9: end for

```

A.2. Query-Time Retrieval

Algorithm 2 IS-RAG Retrieval

Input: Query q , mode $m \in \{\text{text}, \text{cognitive}, \text{hybrid}\}$, top- k , store $\mathcal{S}$, $\alpha=0.4$, $\beta=0.3$

Output: Ranked list of k chunks

```

1:  $(\Sigma_q, \Delta_q) \leftarrow \mathcal{P}(q)$  ▷ detected schemas and domains
2:  $\mathbf{e}_{q,t} \leftarrow \mathcal{E}(q)$ ;  $\mathbf{e}_{q,c} \leftarrow \mathcal{E}(\text{SERIALIZE}(\mathcal{P}(q)))$ 
3: for each chunk  $(C_i, \mathbf{e}_{t,i}, \mathbf{e}_{c,i}, M_i) \in \mathcal{S}$  do
4:    $s_t \leftarrow \cos(\mathbf{e}_{q,t}, \mathbf{e}_{t,i})$ ;  $s_c \leftarrow \cos(\mathbf{e}_{q,c}, \mathbf{e}_{c,i})$ 
5:    $s \leftarrow \begin{cases} s_t & \text{if } m = \text{text} \\ s_c & \text{if } m = \text{cognitive} \\ (s_t + s_c)/2 & \text{if } m = \text{hybrid} \end{cases}$ 
6:    $n \leftarrow |M_i.\text{details}|$ 
7:    $\sigma \leftarrow |\{d \in M_i.\text{details} \mid d.\text{schema} \in \Sigma_q\}| / \max(n, 1)$ 
8:    $\delta \leftarrow |\{d \in M_i.\text{details} \mid d.\text{domain} \in \Delta_q\}| / \max(n, 1)$ 
9:   if  $M_i.\text{schemas} \cap \Sigma_q \neq \emptyset$  then
10:     $\hat{s}_i \leftarrow s \cdot (1 + \alpha \sigma + \beta \delta)$  ▷ Eq. 1
11:   else
12:     $\hat{s}_i \leftarrow s$  ▷ no schema match — no boost
13:   end if
14: end for
15: return  $\text{TOPK}(\{\hat{s}_i\}, k)$ 

```
